

1 Introduction

1.1 Computing Ethics

Computing has a global impact, beyond the introduction stage and into the permeation stage - computers have become part of every aspect of daily life for most people. How can ethics guide us with change? (James) Moor believes we can use *reason & relativity* in computer ethics - we disagree with routine ethics and cultural relativism tech allows us to do new things and raises new ethical questions. One view of computer ethics is : - Routine Ethics - ethical problems are no different in computer as in other fields - Cultural relativism - local customs and laws determine what is right and wrong, and as computing crosses cultural boundaries the problems are intractable

An alternate view (Moor, 1998) is that: - Routine Ethics underestimates the unique ethical challenges that computing technology poses - Cultural relativism underestimates the stability of our core human values - People generally prefer to live and not die - People have a set of common shared set of values, these are relative, reasoned, and can be argued or changed.

1.2 Properties of Computers that Raise Problems

Logical Malleability - Computers can perform any activity that has inputs, outputs and connecting logical operations - infinite possibilities Invisibility Factor - Most of the time and under most conditions computer operations are invisible (3 types: invisible abuse, invisible programming, invisible complex calculation) Informationally enriching - Computer technology is put to use in different fields of activity. Once in place it is used to modify activity and to enhance it with extra knowledge.

1.2.1 Logical Malleability

This makes computer revolutionary - Can be manipulated to do any activity with inputs, outputs, logical operations. Computers are general purpose machines. They can be manipulated: Syntactically - can alter what a computer does by changing its program; or Semantically - can use states of a computer to represent anything one chooses

1.2.2 Invisibility Factor

Invisible abuse - intentional use of the invisible operations of a computer to engage in unethical conduct - i.e. salami slicing case Invisible programming - values embedded into programs by programmers, unintentional or intentional - i.e. google Invisible complex calculation - computers are capable of calculations too complex for easy human inspection - i.e. decision making software

1.2.3 Informationally enriching

Used in diverse activities. Once in place they are used to enhance capabilities and improve performance. Computerised activities often become informationalised. The processing of information becomes a crucial ingredient in the performing and understanding of the activities themselves, so activities become informationally enriched.

1.3 Ways of Approaching Computer Ethics

Ethics is not just a decision about right and wrong. it is a process that can involve a) analysis of the nature and social impact of computer technology b) formulation and justification of policies for the ethical use of technology.

Although we need (a) before (b), we often encounter problems before doing either. Technology innovation moves quickly and is disseminated widely, often without an understanding of unintended consequences, often creating a policy vacuum

1.4 Professional Issues

Computing is not a profession in the same way as medicine, law, engineering. Professional bodies, such as BCS, ACM, IEEE aim to provide education and guidelines to encourage professionalism in practice. Professionals require mastery of a body of knowledge and usually have a good deal of autonomy in their work. They have to take responsibility for their decisions and their actions. How might that affect what you do in the workplace?

2 Crime & Cybersecurity

The essential elements of a crime are; an illegal action or omission (Actus Reus), excluding thoughts but not words; and intention (Mens Rea) - a voluntary or deliberate act. However, some offences have strict liability, meaning you don't have to prove intention. Simply performing the forbidden act is to offend. These are often to do with breaking regulation, and lead to conviction without criminal intention or criminal negligence. **civil, not criminal offences**

2.1 Cyber Dependent Crimes

Cyber crimes fall broadly into two main categories: Illicit intrusions into computer networks, such as hacking; and disruption of computer & network functionality, i.e. DoS attacks Cyber crimes are committed for many different reasons. For example, - Highly skilled individuals/groups who aim to commit crimes - Highly skilled individuals/groups who aim to protest - Low skilled individuals/groups using cyber tools developed by others - Organised criminal groups - Cyber terrorists who intent to cause maximum disruption and impact - States and state sponsored groups launching cyber attacks on other countries - Insiders and (ex-)employees with privileged access to computers and networks

2.2 Computer Misuse Act 1990

Section 1 : unauthorised access to computer material - must have knowledge that you are unauthorised & there must be intention to access the system
Section 2 : unauthorised access with intent to commit further offences - as above with intent to commit a serious further offence, i.e. blackmail, theft
Section 3 : unauthorised acts with intent to impair, or recklessness as to impairing, the operation of a computer. Impair operation of any computer, prevent or hinder access to program or data (i.e. DoS), impair operation of program or reliability of data. Extra sections added 3A (hacking tools), 3ZA (damage to human welfare). The penalties for this are a fine or up to 2(s1), 5(s2 & s3A) or 10 years (s3) or life (s3ZA) in prison

2.2.1 s3A CMA - Hacking Tools

Making, possessing, supplying articles for computer misuse. Prosecutors should be aware that there is a legitimate industry concerned with the security of computer systems that generate 'articles' (this includes any program or data held in electronic form) to test and/or audit hardware and software. Prosecutors need to ascertain that the suspect has a criminal intent. There are some issues to consider: Was the article developed primarily and deliberately for wrongdoing?; Is it available on a wide scale commercial basis and sold legitimately?; Is it widely used for legitimate purposes?; Does it have a substantial installation base?; What was the context in which the article was used compared to the original purpose? An expert witness is often required

Case example - R v Paul McLoughlin McLoughlin, a student, wanted free gaming facilities. He wrapped Is-tealer(*stupid name*), password stealing software, inside several software programs chosen to trick users into downloading them. He harvested logins of over 100 people this way. A US resident complained to the University that passwords to her online accounts had been compromised. Police, working with McAfee and the University of Salford identified the encrypted details of an FTP servers embedded within the malware, incriminated McLoughlin. Charged with making, supplying or obtaining articles for use in offence under section 1 or 3. Given 8 months, suspended for 12. First conviction for s3A offence.

2.2.2 s3ZA CMA - damage to human welfare

Causing or risk of causing serious damage to human welfare, environment, economy, national security. Designed to cater for computer misuse, where the impact is to cause damage to for example, critical national infrastructure and where a higher penalty should be given. Here the definition of "critical national infrastructure" could be understood to be an asset or system which is essential for the maintenance of vital societal functions, health, safety, security, economic, or social well-being of people, i.e. power plants, transport networks, or governments. The disruption or destruction of which would have a significant impact

2.2.3 CMA Prosecution - Factors

Does the institution, company, or other body have in place robust and up to date contracts, terms and conditions, or acceptable use policies? Are employees, students, customers, and others made aware of the CMA and what is lawful and unlawful? Do employees, students, customers, or others have

to sign a declaration that they do not intend to contravene the CMA?

2.3 Link to Data Protection Act 2018

DPA 2018 creates a number of offences in relation to the control and access to data - section 144 : An offence for a person to intentionally or recklessly make a false statement in response to an information notice. Section 170- An offence to: Obtain or disclose personal data without the consent of the data controller; procure the disclosure of personal data to another person without the consent of the controller; or after obtaining personal data, to retain it without the consent of the person who was the controller in relation to the personal data when it was obtained. Case- Kim Doyle working for RAC prosecuted under CMA for accessing & passing on personal data about individuals who has accidents

2.4 The Ethics of Hacking

Hacking is conducting technical activities with the intent of exploiting vulnerabilities within a computer system, network or firewall to obtain unauthorised access. *black-hat hackers* - unauthorised, malicious hackers *grey-hat hackers* - exploit security vulnerabilities to spread public awareness that the vulnerability exists *white-hat hackers* - authorise, ethical hackers, are generally hired to test systems using a range of methods to identify security weaknesses. Have a code of ethics which prohibits sharing how they breached security with anyone outside the client organisation

2.4.1 DPA/UK GDPR & Cybersecurity

The DPA/UK GDPR requires that personal data must be processed securely using appropriate technical and organisational measures. It does not mandate a specific set of cyber security techniques but expect companies to take 'appropriate' action. This means identifying and managing risk. What is appropriate for an organisation will depend on circumstances, as well as the data being processed and therefore the risks posed.

2.4.2 Cybersecurity Accountability

DPA and GDPR focus on explicit accountability for data protection, placing a direct responsibility on companies to prove they comply with the principles of the regulation. Companies need to commit to mandatory activities such as staff training, internal data audits and keeping detailed documentation if they wish to avoid falling foul of the DPA & GDPR rules. Malign staff actors cant be prosecuted if the authorisation isn't clear. Personal data breaches must be reported to the information commissioners office without undue delay (if it meets the threshold for reporting) and within 72 hours

2.4.3 National Cyber Security Centre

Launched in 2016 NCSC provides a service for public and private sector organisations in relation to cyber security, it is a knowledge hub providing practical guidelines to the public. Responds to cyber security incidents and works with organisations and the government to reduce harm caused. Uses industry and academic expertise to improve the UK's cyber security capability. Reduces risks to the UK by securing public and private sector networks

3 Values & Virtue: Ethics in Action

Ethics is concerned with what is good for individuals and society, i.e.: moral decisions, what is good and bad?; our responsibilities and rights; how to live a good life; a branch of moral philosophy. Ethical behaviour is not always: what the law tells you to do; decided by the majority; determined by authority.

3.1 How do we make ethical decisions?

****moral realism**** - moral principles have an objective foundation, and are not based on subjective human convention
****moral subjectivism**** - moral judgements reflect personal preference or opinion
****moral relativism**** - moral standards are grounded in social approval, and therefore vary: individually (individuals make their own decisions) or socially (societies make their own decisions and change over time)

3.1.1 Ethical Approach - Consequentialism

Of all the things a person might do at any given moment the morally right action is the one with the best overall consequences. There are two principles: whether an act is right or wrong depends only on the results of that act; and the more good consequences an act produces the better or more morally right that act is consequentialism gives us this guidance when faced with a moral dilemma, a person should choose the action that maximises good consequences and it gives this guidance on life, people should live so as to maximise good consequences.

****utilitarianism**** - people should maximise human welfare or well-being which used to be called 'utility'- Jeremy Bentham & John Stuart Mill
****ethical altruism**** - people should maximise the welfare of others - Auguste Comte
****hedonism**** - an early form - people should maximise pleasure

Practice of Consequentialism It is difficult to assess the ethical of every single act (act consequentialism). Consequentialism produces this conclusion: "no type of act is *inherently* wrong, not even murder, it depends on the result of the act". However, we can derive ethical rules by considering the general consequences of types of acts (rule consequentialism) i.e. murder is wrong because it deprives individuals of their life and produces other bad consequences ****Act consequentialism**** looks at every single moral choice anew, so, a particular action is morally action is only good if it produces more overall good than any alternative action ****rule consequentialism**** bases moral rules on their consequences, so whether acts are good or bad depends on moral rules, which are chosen solely on the bases if their consequences. Individuals should apply rules to make a moral choice. the rules are those that would produce the best results if adopted by the most people.

3.1.2 Ethical Approach - Deontology (duty based ethics)

Deontological ethics are concerned with what people do, not with the consequences of their actions. i.e. do the right thing, do it because its the right thing to do, don't do the wrong things, avoid them because they are wrong. In this approach, you don't justify an action by showing that it produces good consequences (non-consequentialist). some kinds of action are wrong or right in themselves regardless of consequence.

Deontologists live in a universe of moral rules such as: it is wrong to kill; it is wrong to steal; it is wrong to tell lies; it is right to keep promises. Someone who follows duty based ethics should do the right thing even if it produces more harm or less less good than going the wrong thing - people have a duty to do the right thing even if it produces a bad result

****Kant's categorical Imperative**** - Categorical imperative (a rule that is true in all circumstances) - basis of all rules there are two ways of expressing it, but its one rule: always act in such a way that you would be willing for it to become a general law that everyone else should do the same in the same situation; and act so that you treat humanity, both yourself and others, always as an end and never merely as a means to an end

3.1.3 Ethical Approach - Virtue Ethics

A right act is the action a virtuous person would do in the same circumstances. Person based rather than action based, about moral character. Virtue is an excellent character trait that is whole heartedly understood by the person who possesses it. We acquire virtue through practice and by honing virtuous habits people will make better choices when faced with ethical challenges. Virtue ethics provides guidance about the characteristics of a good person

****justice**** - Justice requires us to treat human beings equally and impartially
****fidelity**** - Fidelity requires that we treat people close to us with special care
****self-care**** - we have a unique responsibility to care for ourselves mentally physically & spiritually
****prudence**** - all of the above and looks for opportunities to acquire more virtue

3.1.4 Ethical Approach - Moral Relativism

Moral relativism states that there are no objective moral truths. there are no objective moral facts, therefore 'killing people is wrong' can't be objectively true. Many forms of relativism go a bit further and say that moral statements describe how the speaker feels about a particular ethical issue. Moral statements are just factual statements about the attitude the speaker holds on a particular issue, i.e. if i say "lying is wrong" → i disapprove of telling lies

Some moral relativists are descriptive; describing different cultures - different cultures take different moral stances, at any one time and over time; and describing individuals - different individuals take different moral views based on their beliefs.

Some moral relativists are normative - morality has no objective factual basis, therefore it is not possible to say that one moral stance is right and another is wrong.

3.2 Values

****ethics**** looks at moral principles and guidelines that help people think about what is right and wrong. ethics seeks to answer questions about how individuals should act, make decisions, and interact with others.

****values**** reflect beliefs about what individuals consider important, desirable, and meaningful in life. Values serve as motivators that influence attitudes, behaviours, and choices.

Shwartz, a psychologist, developed a theory of values based on many empirical studies across different cultures over years. Ten basic human values form a circular continuum, defined according to the motivation that underlies each of them. Values refer to desirable goals that motivate action. Values guide

the selection of action, policies, people, and events. The relative importance of multiple values guides action. Schwartz's 10 Basic Human Values reflect universal human motivations. They include **self-direction** (independence and creativity), **stimulation** (excitement and novelty), and **hedonism** (pleasure and enjoyment). Other values focus on **achievement** (success and competence), **power** (control and status), and **security** (safety and stability). Additionally, there's **conformity** (adherence to norms), **tradition** (respect for customs), **benevolence** (concern for close others), and **universalism** (appreciation for all people and nature). These values highlight the tensions between openness and conservation, and self-enhancement and self-transcendence.

4 Risk & Responsibility

Rise of harmful and illegal content online has led to concern. Through social media, users are now interacting a lot more with each other, which opens them to harmful encounters such as trolling, abuse, and the sharing of offensive or illegal material. There are concerns that children and young people are also experiencing this type of harm, putting many in danger of being exposed to illegal and harmful content. On the other hand, many believe that free speech is an important tenet to protect in online spaces

4.1 Online Safety Act 2023

Aims to protect children and adults online, creates a set of duties for social media companies and search services, making them more responsible for their users safety on their platforms. gives providers duties to implement systems and processes to reduce risks their services are used to illegal activity, and to take down illegal content. the strongest protections are designed for children. platforms will be required to prevent children from accessing harmful and age inappropriate content and provide parents and children with clear and accessible ways to report problems online when they do arise. Also protects adult users, requiring major platforms to be transparent about which kinds of potentially harmful content they allow, and give people control over the types of content they want to see.

Criminal offences in OSA 2023 - The criminal offences introduced by the Act came into effect on 31 Jan 2024, covering: encouraging or assisting self harm, cyberflashing, sending false information intended to cause non-trivial harm, threatening communications, intimate image abuse, epilepsy trolling **Protections in OSA 2023** - Requires online services to: take robust action against illegal content and activity, i.e. child abuse, controlling behaviour, sexual violence, etc; Take robust action against content that is harmful to children; enforce age limits consistently and protect children; offer adult users tools to give them control over content they see; rapidly remove suicide/self harm content

4.1.1 OFCOM

the independent regulators for the OSA. It sets out steps providers can take to fulfil their safety duties in codes of practice and advice. it has a broad range of powers to assess and enforce providers compliance with the framework, from running audits to fining companies. providers safety duties are proportionate to factors including the risk of harm to individuals, and the size and capacity of each provider. This makes

sure that while safety measures need to be put in place across the board, small services with limited functionality dont need to take the same actions as large corporations. Ofcom is required to take the users rights into account when setting out steps to take and providers have duties to take regard of users rights when fulfilling their safety duties.

Companies need to work with Ofcom to comply with OSA. providers of online user-to-user services will be asked to: - take proportional steps to prevent users encountering illegal content; manage the risk of offences taking place through the services; - be able to do age verification if necessary; - undertake a risk assessment and manage risks identified; - remove illegal content when aware of it; - protect children from harmful content; - explain the approach taken in terms of service. More stringent measures for larger services, less so for smaller ones

Benefits and Criticisms of the OSA **Benefits** : Child protection - enforces age checking measures to prevent children from accessing age inappropriate content on social media platforms; Content removal - requires tech companies to promptly remove illegal content like CSAM, hate speech, self harm promoting; User control - allows users to filter out unwanted content like abuse and customise their online experience; Transparency and risk assessment - forces large platforms to public risk assessments outlining potential dangers to users, particularly children; Accountability for platforms - holds tech companies responsible for enforcing their own terms of service and taking action against harmful content; Improved reporting mechanisms - provides clear pathways for users to report harmful content and concerns; Enforcement by Ofcom - grants Ofcom power to investigate and sanction companies that fail to comply with safety rules

Criticisms : Overreach - could force platforms to remove too much content, potentially impacting legitimate discussion and viewpoints, due to broad definitions of harmful content/ not removing enough content or doing it quickly enough; privacy concerns - requirement for platforms to scan private messages to detect illegal content with fears of government surveillance; Free speech implication - could stifle free speech by causing platforms to err on the side of caution and remove content that may be controversial but not illegal; Potential for abuse - could be used to target specific groups or viewpoints, leading to censorship; Lack of focus on system design - focuses on content removal and not proactive measures to prevent harmful content from appearing in the first place; Unclear definitions - some provisions in the act lack clear definition, which could lead to inconsistent application and interpretation.

4.2 Freedom of Expression

FoE is protected by Article 10 of the UK Human Rights Act 1998; "Everyone has the right to freedom of expression ... to hold opinions and to receive and impart information and ideas without interference by public authority ... " Restrictions on this include; Protecting others - rights and reputations of others or preventing disorder or crime; Preventing hate speech - including incitement to racial or religious hatred, or terrorism; Protecting confidential information- disclosure of information received in confidence; Maintaining judicial impartiality - the authority and impartiality of the judiciary; Religious freedom - can be limited to protect the rights and freedoms of others; Anti-protest legislation - anti-protest legislation has been passed, with questionable legality.

4.2.1 Defamation

Defamation is (a) to lower a person in the estimation of right thinking members of society generally; or (b) to expose a person to hatred, ridicule, or contempt; or (c) to disparage a person in his or her office, profession, calling, trade, or business. Protecting reputation is important, as reputation is a civil right of the same standing as any other, e.g. right to enjoyment of life. This is covered by the Defamation Act 2013. Defamation is a civil wrong, or tort. Claims generally have to be issues in high court, only sometimes in the county court.

In UK law, libel and slander are types of defamation, which protect a person or organisations reputation from false statements: **libel** - defamatory statements that are written or broadcast, such as in articles, blogs, social media posts, or other written content; **slander** - defamatory statements that are not published and transient, i.e. conversation, impressions, other transient media that are not stored.

a statement is defamatory if it is: potentially and actually defamatory; clearly about the claimant; and was published or communicated. Claimants must prove that the publication of a statement caused or is likely to cause serious harm to their reputation. For companies trading for profit, the harm must also cause or be likely to cause serious financial loss. A government body cant be defamed.

Defences: The defamation act outlines defences for defamation: Truth - A statement can be fully defended against defamation if it is substantially true; Honest opinion - A statement can be defended if it is an opinion based on accurate, referenced facts; Public interest - A statement can be protected if it is on a matter of public interest and the defendant reasonably believed publication was necessary; Operators of websites - If complied with procedures and remove posts when a complaint received ; Peer reviews statements in academic papers; Reports protected by privilege, i.e. court reports

5 Professionalism in Practice

computing is not a profession in the same way as medicine, law, engineering, but there are professional societies(BCS, ACM, IEEE) that try to encourage professionalism in practice. Technology is a powerful tool that needs to be developed with care.

Professionals require mastery of a body of knowledge, usually through higher education, they often have a good deal of autonomy in their work, and have a professional organisation, often acknowledged by government that controls admission and sets guidelines for practice. Professionals fulfil an important social function, doctors, lawyers, etc

Computer professionals possess some of the characteristics(mastered a body of knowledge, usually gone through higher education, many have a degree of autonomy). While there are professional computing bodies there is no legal body in charge of admissions or standards in the field of computing. Computing is crucial to society, but it is not a good in itself as are health and justice. it supports activities in society, but they are varied.

It could be argued that professional function in a special context (Johnson 1985). Typically, that includes relationships with employers, clients, other professionals, and the public. They operate under legal, political, and economic constraints. They have skills that enable them to affect the world. They usually use their skills to produce products or provide services

that are widely used - this usually involves working with a range of others, because of this they are seen as bearing a responsibility to not do harm to individuals or the public.

A professional Code of Ethics serves several functions: Symbolises the professionalism of the group; Defines and promotes a standard for external relations with clients and employers; protects the groups interests; Codifies members rights; Expresses ideals to aspire to; Offers guidelines in grey areas.

5.1 Codes of Conduct

BCS CODE OF CONDUCT - Applies to all members of BCS, and covers 4 main areas: **Public Interest** - Have regard for public health, privacy, security, and wellbeing of others and the environment. Have regard for the legitimate interests of third parties. Don't discriminate against others in any of your dealings. Promote equal access to IT and generally support inclusion. **Professional Competence & Integrity** - Only undertake work within your competence. Don't claim level of competence you do not possess. Keep professional knowledge up-to-date. Know relevant legislation. Respect the viewpoints of others. Avoid injuring others, property, reputation, employment. Reject bribery. **Duty to Relevant Authority** - Carry out work with due care and diligence. Avoid situations which may result in a conflict of interest between you and your relevant authority(company, client, etc). Accept professional responsibility for your work and the work of colleagues you supervise. Don't disclose or use for personal gain confidential information relating to your work, except with permission or if legally required. Don't misrepresent or withhold information about the performance of products. **Duty to the Person** - Accept personal duty to uphold reputation of the profession. Improve professional standards by taking part in their development, use and enforcement. Uphold the reputation of the BCS. Act with integrity and respect in your relationships with others. Encourage and support members in their professional development.

ACM CODE OF ETHICS & PROFESSIONAL RESPONSIBILITIES **Code of Ethics** - 1. Contribute to society and to human wellbeing, acknowledging that all people are stakeholders in computing. 2. Avoid harm. 3. Be honest and trustworthy. 4. Be fair and take action not to discriminate. 5. Respect the work required to produce new ideas, inventions, creative works, and computing artifacts. 6. Respect privacy. 7. Honour confidentiality. **Professional Responsibilities** - Strive to achieve high quality work. Maintain high standards of competence, conduct & ethics. Know and respect rules pertaining to your work. Accept and provide appropriate professional review. Give comprehensive evaluation of computer systems incl. risks. Only perform work in areas of competence. Foster public awareness and understanding of computing. Only access computing resources when authorised. Design and implement systems that are secure **Professional Leadership Principles** - Ensure the public good is the central concern. Encourage the fulfilment of social responsibilities by all colleagues. Manage people and resources to enhance quality of working life. Support policies that reflect this code. Create opportunities for others to grow as professionals. Use care when modifying or retiring systems. Recognise and take care with systems that are integrated into the infrastructure of society.

Characteristics, Problems, and Application of Codes They are not simple ethical algorithms that generate ethical

decisions. Sometimes elements of the code may be in tension with each other or other sources. Requires the IT professional to use ethical judgement to act in the spirit of the code of ethics. A good code of ethics/conduct will set out fundamental principles that require thought rather than blind allegiance.

Codes don't cover every case, nor should they. There is little penalty for non compliance. Realistically compliance requires a personal code of ethics that is broadly in line with the professional code.

Apply codes by: Consider the ethical dilemma you are facing; Decide which principles and rules are at stake; Decide if there are any conflicts between principles in this case; Make a decision based on prioritisation of principles

Equality, Diversity & Inclusion (EDI) is related to professionalism and relevant for the IT workplace an IT systems that we develop as software engineers. There are compelling reasons for considering this in SE development: Legal issues - i.e. not to discriminate; Professional issues - mentioned in the codes of conduct; Ethical issues - fairness and justice; Social issues - insuring that IT systems represent and can be useful to all of society. The Equality Act 2010 protects 9 characteristics - age, disability, gender reassignment, marriage or civil partnership, pregnancy and maternity, race, religion or belief, sex, sexual orientation.

6 Data & Decisions

Data Protection Act DPA first passed in 1984 to protect personal data (PD) in paper based filing systems and on computers. Updated in 1998 in response to new EU law and advances in tech, more detailed, concerned with how PD was processed. in 2002 DPA was supplemented with EU PECR, which focused on email marketing and website consent for cookie, known as cookie law. in May 2018 EU enacted General Data Protection Regulation (GDPR) some sections left open to interpretation, DPA updated to supplement GDPR. in 2020, uk left eu, keeping GDPR with small changes, enacted as UK GDPR, i.e. age of child consent is 13 in uk, 16 in eu, changes to how PD about criminals is handled, and how PD is used for automated decision making. Organisations have to comply with UK GDPR and DPA if working in UK, if they process data of EU residents they must comply with EU GDPR

6.1 UK Data Protection Law

DP legislation sets out rules for the processing of information about living identifiable individuals, or data subject, by organisations, or data controllers. Applies to organisations in public and private sectors. it applies to electronic and most paper records. doesn't apply to anonymous information or to information about the deceased. Since 1 Jan 2021, the principle legislation has been the UK GDPR and the DPA 2018 which supplements the GDPR

PD is information that relates to an identified or identifiable living individual. This means anything about a person, name, address, phone no., email address, bank acc no, NI no, student id, library ca. Also, its personal if it can be used to identify a person when pieced together with other data, i.e. IP address or username for website, data confirming physical presence somewhere, fingerprints, CCTV, GPS location.

If you cannot directly identify an individual from that information, then you still need to consider whether the individual is identifiable. When considering whether information relates

to an individual, you need to take into account a range of factors, including the content, the purpose for which you are processing it, and the likely impact or effect of that processing on the individual. Information which has identifiers removed or replaced in order to pseudonymise the data is still personal data for the purposes of GDPR. Information which is truly anonymous is not covered by the GDPR. If information that seems to relate to a particular individual is inaccurate it is still personal data, as it relates to that individual.

Stronger protections for type of data that might be used to discriminate about people - sensitive personal data or special category data. Need explicit permission or a very good reason: Race; Ethnic background; sexual orientation & sex life; Religious beliefs; Political opinions; Trade Union membership; Health data med history, including data by fitness apps; Biometrics if use for ID; Genetic information.

The 7 Data protection principles are the following: 1. Processed lawfully fairly and transparently - only if legal basis, i.e. consent, contract, legal obligation, vital interests, public interest, legitimate reason. 2. Processed only for specified explicit and legitimate purposes - tell people what you will do, even if sharing or selling data with others. 3. Adequate, relevant, and limited to only what is necessary - minimum amount of data. 4. Accurate and, where necessary, kept up to date. 5. Not kept for longer than necessary - DC must establish time limits for keeping PD. 6. Must be kept securely, preserving confidentiality, integrity, and availability - this means keeping PD encrypted if stored, employees also need to know this. 7. Data controller must be accountable - must keep detailed records as evidence of policies being followed, third parties must also follow principles.

6.2 Data Controllers (DC) and Data Processors

The UK GDP defines a data controller (DC) as "the natural or legal person, public authority, agency or other body which, alone or jointly with others, determines the purposes and means of the processing of personal data". Controllers make decisions about processing activities. They control PD being processed and are responsible for it. Can be a company or other legal entity, i.e. public authority, or an individual, i.e. sole trade or self employed professional. Individuals processing PD for household activity not subject to DP law.

The UK GDPR defines a processor as "means a natural or legal person, public authority, agency or other body which processes personal data on behalf of the controller". These act on behalf of a DC and under the DC's authority. They serve the controller's interests and not their own. May make day-to-day decisions but should only process personal data in line with the controller instructions. If they act without the DC's instructions, they become a controller for that activity and have the same liability as a DC

6.3 Information Commissioner's Office (ICO)

ICO is the UK's independent regulator for data protection and freedom of information, their role is to: uphold the UK GDPR and DPA. Uphold the Freedom of Information Act. Uphold information rights of individuals and organisations. Monitor and investigate breaches. Take enforcement action. Promote good practice. Support organisations.

Data Access Rights : To be informed of how personal data is being used - privacy notices. Access to personal data -

known as making a subject access request. Rectification of inaccurate personal data. Erasure of personal data where appropriate - right to be forgotten. Restrict processing of personal data pending its verification or correction. Data portability - to receive copies of their personal data in a machine readable format. To not be subject to automated decision making - profiling, direct marketing, for research. Objecting to the way their personal data is being used.

Response to a rights request normally should be sent within one month, but nearly all rights are qualified in various ways and there are exemptions.

An important aspect of complying with data protection law legislation is being open and transparent with individuals about how their personal data is being used. The supply of this information - through documents variously known as privacy notices, data protection statements, data collection notices, privacy policies, is essential. It takes place in numerous targeted ways depending on the context of the interaction with the individual. It is the responsibility of the DC to provide notices.

Data Subjects have the right to access a copy of personal data, should be provided in a month and free of charge, easy to access and understand. If complex, can take up to 3 months and a small fee may be charged, by the request should still be responded to within a month with an explanation for why it will take time. Individuals have the rights to : receive a copy of your personal data in a structured, commonly used computer readable format so you can pass it on; ask a DC to transmit your data directly to another DC, that means to provide electronic data in an open format, e.g. CSV, XML, JSON; Only include raw data that you gave, (name, address, etc.) DCs are not obliged to pass on data they've inferred from your data, i.e. likes and dislikes; make a data access request.

Automated decision making is increasingly common. subjects have the right to not be subject to decision made purely by automated means. Only allowed to do if explicit consent or authorised by law. If using sensitive data must have your explicit consent or for a public interest. Good practice for DC to tell you details of any profiling or automated decision making they do, including where the data comes from. DC must explain how you can access data used to create your profile and simple ways to request human intervention or to challenge an automated decision. Any DC who falls foul of DP rules will be held to account. Breaches leading to disclosure, loss, destruction or alteration of PD that present a serious risk to anyone's rights and freedoms should be reported to ICO within 72 hours and the subject should be notified asap. Actions by ICO on orgs who fail to comply include warnings, temporary bans on data processing, if very serious, fines up to 17.5M or 4% of annual turnover in UK.

7 Society & Surveillance

7.1 Human Rights

United Nations - created in 1945 by representatives of 50 countries, a response to the end of WW2. The Commission of Human Rights set up by the UN to draft the Universal Declaration of Human Rights (UDHR). It was made up of 18 members from various political, cultural, and religious backgrounds, chaired by Eleanor Roosevelt.

This was a roadmap to guarantee the rights of every individual everywhere. UDHR adopted by UN general assembly, consisting of over 50 member states, on 10 Dec 1948, with 8

nations abstaining but none dissenting. it marked a historical turning point, as a text promoting peace and diplomacy.

there are 30 basic human rights: 1. have and respect human rights. 2. life. 3. Freedom from torture. 4. Freedom from slavery. 5. Liberty and Security. 6. Seen as a person. 7. Equality. 8. Remedy in Law. 9. Freedom from arbitrary arrest. 10. Fair trial. 11. Innocent until proved guilty 12. Privacy & family life 13. Freedom of movement 14. Freedom from persecution 15. A nationality 16. To marry 17. Own property 18. Freedom of thought, conscience, religion 19. Freedom of opinion & expression 20. Freedom of association 21. Take part in govt and vote 22. Human dignity 23. To work & receive fair pay 24. To rest & leisure 25. Adequate standard of living 26. Education 27. Participate in cultural activities 28. To have rights realised internationally 29. Duty to community 30. Not to have any rights removed.

The European Convention on Human Rights (ECHR) protects the human rights of people in countries that belong to the Council of Europe, including the UK. The Council of Europe was founded after world war 2 to protect human rights and the rule of law and to promote democracy. The convention was based on the UDHR, it was signed in Rome in 1960 and came into force in 1953. Established in the European Court of Human Rights. Judgements finding violations are binding on the states concerned and they are obliged to carry them out. Post Brexit some UK politicians want to leave the ECHR

7.2 Human Rights Acts 1998

The UK Act received royal assent in 1998 and came into force on 2 Oct 2000. Its aim is to give further effect in UK law to the rights contained in the ECHR. The Act gives UK courts a remedy for breach of a Convention right, without the need to go to the European Court of Human Rights. Some additions to UDHR are added: Protocol 1, Article 1 - Right to peaceful enjoyment of your property; Protocol 1, Article 3 - Right to participate in free election; Protocol 13, Article 1 - Abolition of the death penalty.

7.3 Surveillance and GDPR/DPA

****Surveillance technologies**** : Closed Circuit Television Systems (CCTV); Automatic Number Plate Recognition (ANPR); Body Worn Video (BWV); Unmanned Aerial Vehicles (UAVs, or drones); Facial Recognition Technology (FRT), Commercial products such as smart doorbells and surveillance in vehicles; Workplace monitoring, live-streaming; Other commercially available surveillance systems that have the potential to process personal data.

DCs must implement appropriate technical and organisational measures & identify lawful basis. If a surveillance system processes PD must notify and pay a data protection fee to ICO unless exempt. Must have appropriate measure and records in place to be able to demonstrate compliance. Maintain a record of the processing activities (article 30), including purpose(s), data sharing agreements, retention periods. Take a data protection by design and default approach and perform a Data Protection Impact Assessment (DPIA) for processing likely to result in high risk to individuals, including processing special category data, monitoring publicly accessible places on a large scale, or monitoring individuals at a workplace. DCs must also assess whether surveillance is appropriate considering the reasonable expectation of the individuals affected and the potential impact on their rights and

freedoms. Record considerations in a DPIA prior to deployment. If high risks cannot be mitigated, prior consultation with the ICO is required

7.3.1 FRT

FRT identifies a person from a digital facial image and analyses facial features to produce a biometric template. Depending on use FRT involves processing PD, biometric data, and often special personal data. Under UK GDPR processing biometric data to uniquely identify an individual is prohibited unless a lawful basis under article 6 (consent, contract, legal obligation, vital interest, public interest, legitimate reason). And a condition in article 9 can be satisfied, these are: explicit consent; employment, social security and social protection law; vital interests; not for profit bodies; made public by the data subject; legal claims and judicial acts; substantial public interest conditions; health or social care; public health; archiving, research and statistics.

FRT in public spaces is where data is scanned and checked against a watch list, if found usually human decision is made before action. If system is completely automated higher standard is needed. If private company, consent and option to opt out is important. To be accountable when using FRT, must be able to explain: the lawful basis; why FRT is necessary in the circumstances or in the public interest; why less intrusive options rules out; assessment of the likelihood that objectives of the FRT will be met; how effectiveness has been measured.

the ICO made the following FRT checklist: Conducted a Data Protection Impact Assessment (DPIA) addressing need to FRT, lawful basis for its use and explores the impacts on the rights and freedoms of individuals whose personal data are captured for every deployment. Document our justification for FRT, and the decision-making behind justifications. A sufficient volume and variety of training data has been included to assist accurate performance. Chosen an appropriate resolution for the cameras we use and tested equipment. Positioned cameras in areas with sufficient lighting, to ensure good quality images are taken. Can clearly identify false matches, and true matches. Can record false positive or false negative rates where appropriate. Can amend to correct false positive or false negative rates that are too high. Watchlists used are constructed in a way compliant with data protection law. Considered if Equalities Impact Assessment (EIA) is required to meet Equalities Act 2010. Complied with the Surveillance Camera code of practice where required.

Digital technology has a double role as a utility and entertainment, it is part of the social fabric of our lives. Technology pervades our workplace, education, homes, and environment infrastructure. No one knows the social, psychological, and intellectual implications of such a reliance on technology. Predictions abound that it will enhance intelligence and performance and change behaviour, others think that it diminishes some skills and attributes. Beliefs about the values and perils are not new. Since the beginning of industrialisation technology has been both valued and feared as a force of change.

8 Intellectual Property Rights(IPR)

legally protected ability to control ownership. Protection for intangible but valuable things, i.e. writings, inventions, music, drawing, software - things that can be duplicated easily. Protected by copyright and patent law. Allows creators to benefit from their ideas but avoid monopoly

8.1 Copyright

Need permission to perform restricted acts(copying, publishing, adapting or translating). Key statutes are the Copyright, Designs and Patents Act 1988, and The Copyright (Computer Programs) Regulations 1992. Copyright protects original works created by the author, i.e. literary, dramatic, music & artistic works, sound recordings, films, and broadcasts, Typographical arrangements of published editions, including layouts and fonts, software, software design materials, and databases. It is not meant to protect ideas themselves, but includes detailed plot of story as well as words. Its easy to get copyright, hard to get a patent.

Copyright exists from the creation of the work. No special action is necessary, copyright asserts a claim, but is not necessary. The author of the work usually owns the copyright. A work created during employment is owned by the employer not the author unless agreed otherwise. Copyright lasts 70 years after authors death for written works, and 70 years from creation/publication for film/sound, and varies for other types of work.

Copyright doesn't protect ideas, procedures, methods or operations, or mathematical concepts. Copyright protects the expression of an idea, something has to be formulated and written down or recorded. Judges have spent a long time deciding whether or not something is an idea or an expression. (Idea - an attractive spy saves the world, Expression - Skyfall, Spectre - text and plot are protected)

Under CPDA Sect 28 - 76, there are the following permitted acts: Fair dealing, which is non commercial research review or criticism, photocopies of work for students, or studying a program to see what it does; Time-Shifting; Taking back-up copies of computer programs; Corrections of errors in software; Decompiling parts of code if no other way to obtain information needed to interface to a program - mustn't use to create a similar program.

It is not an infringement of copyright for a lawful user of a copy of a computer program to observe, study, or test the functioning if the program. Nor is it to determine the ideas and principles and which underlie any element of the program.

8.2 Patent

A monopoly right to stop others making, using, or selling an invention, even if they think of it for themselves. Lasts up to 20 years, after which it is passed into the public domain. The right can be sold or licensed to others, and can have value. The invention must be disclosed.

UK Statutes: Patents Act 2004, Patents Act 1977, CDPA 1988, Patents Designs and Marks Act 1986. Regulations/Statutory Instruments: Patents Rules (various years).

Inventions which are products or processes with new functional or technical aspects are covered by patents, patents describe how things work, what they do, how they do it, what they are made of, or how they are made. The invention must also be new, involve an inventive step, be capable of industrial application, and must not be excluded.

Patent Exclusions include: A discovery, scientific theory or mathematical method; A literary, dramatic, musical, or artistic work, or any other aesthetic creation whatsoever; A scheme, rule, or method for performing a mental act, playing a game or doing business, or a *computer program*; The presentation of information; These exclusions apply both in UK law (Patents Act 1977) and in EU law (European Patents Convention Art. 52) But this only prevents things from being

treated as inventions only to the extent that a patent relates to that thing as such

8.3 Laws of Confidence

A civil tort based on common law not statute law, imposes a duty of confidentiality. This can be used to protect new ideas before a contract is written, they are expressed in a copyrightable form, or a patent is granted. Only for confidential information, if information is published legitimately, the duty of confidentiality disappears.

Confidentiality protects any information that is not trivial or gossip, nor easily available by other means. Information need not be completely secret or highly sensitive, others can know as long as it is confidential. Covers wider scope than patents or copyrights - covers trade secrets, client lists, personal information, algorithms, intentions, and plans;

Something is a breach of confidence if information has a necessary quality of confidence, i.e. something worth keeping secret. Information was shared in circumstances giving an obligation of confidence, like specific conditions statement (sometimes in a contract), or obvious to reasonable person. There must be an unauthorised use or disclosure of the information with a risk of damage to originator of information.

Public Interest Defence can be taken where disclosure is of significant benefit to society, although this is rare, applies to information for police, taxman, or to public where important. relates to Public Interest Disclosure Act 1998 - protect workers revealing information about past present or future malpractice in the workplace, of the right type (i.e. a qualifying disclosure) excluding official secrets or disclosure to lawyer, and to the right person and in the right way.

Employees cannot disclose employer's confidential information, have obligations of confidence and good faith. After leaving employment, can disclose and use general skills and knowledge acquired, but cannot disclose highly confidential information learnt while employed. Contract of employment can enforce confidentiality during employment and can prevent an ex-employee soliciting customers and working in a competing business (may be subject to reasonableness)

Advantages: Prevents ideas being stolen, protects before other protection in place. Balanced protection for employees/employers Disadvantages: Unethical decisions may be hidden (protection with the Public Interest Disclosure Act 1998). No time limit, so society can't benefit from improvements (compared with copyright/patent)

There are different types of software licences: Proprietary Licences - source code considered a trade secret and IPR held by publisher; FOSS (also public domain) - free to use, change, copy, study; Permission licences - OS with minimal restrictions usually related to modifications/distribution; Copyleft - OS allowing derivative work but require it to be OS & freely available